ideaspark

ESP32 2.8 inch LCD Board

Getting Started with the Arduino IDE · R1.0

English version — Click here to access the file: https://github.com/GJKJ/ESP3228LCD

Versión en español — Haga clic aquí para acceder al archivo: https://github.com/GJKJ/ESP3228LCD

النسخة العربية — انقر هنا للوصول إلى الملف: https://github.com/GJKJ/ESP3228LCD

Versão em português — Clique aqui para acessar o arquivo: https://github.com/GJKJ/ESP3228LCD

Русская версия — Нажмите здесь, чтобы перейти к файлу: https://github.com/GJKJ/ESP3228LCD

日本語版 — ファイルはこちらをクリックしてください: https://github.com/GJKJ/ESP3228LCD

Version française — Cliquez ici pour accéder au fichier : https://github.com/GJKJ/ESP3228LCD

Deutsche Version — Klicken Sie hier, um auf die Datei zuzugreifen: https://github.com/GJKJ/ESP3228LCD

한국어 버전 — 파일을 보려면 여기를 클릭하세요: https://github.com/GJKJ/ESP3228LCD

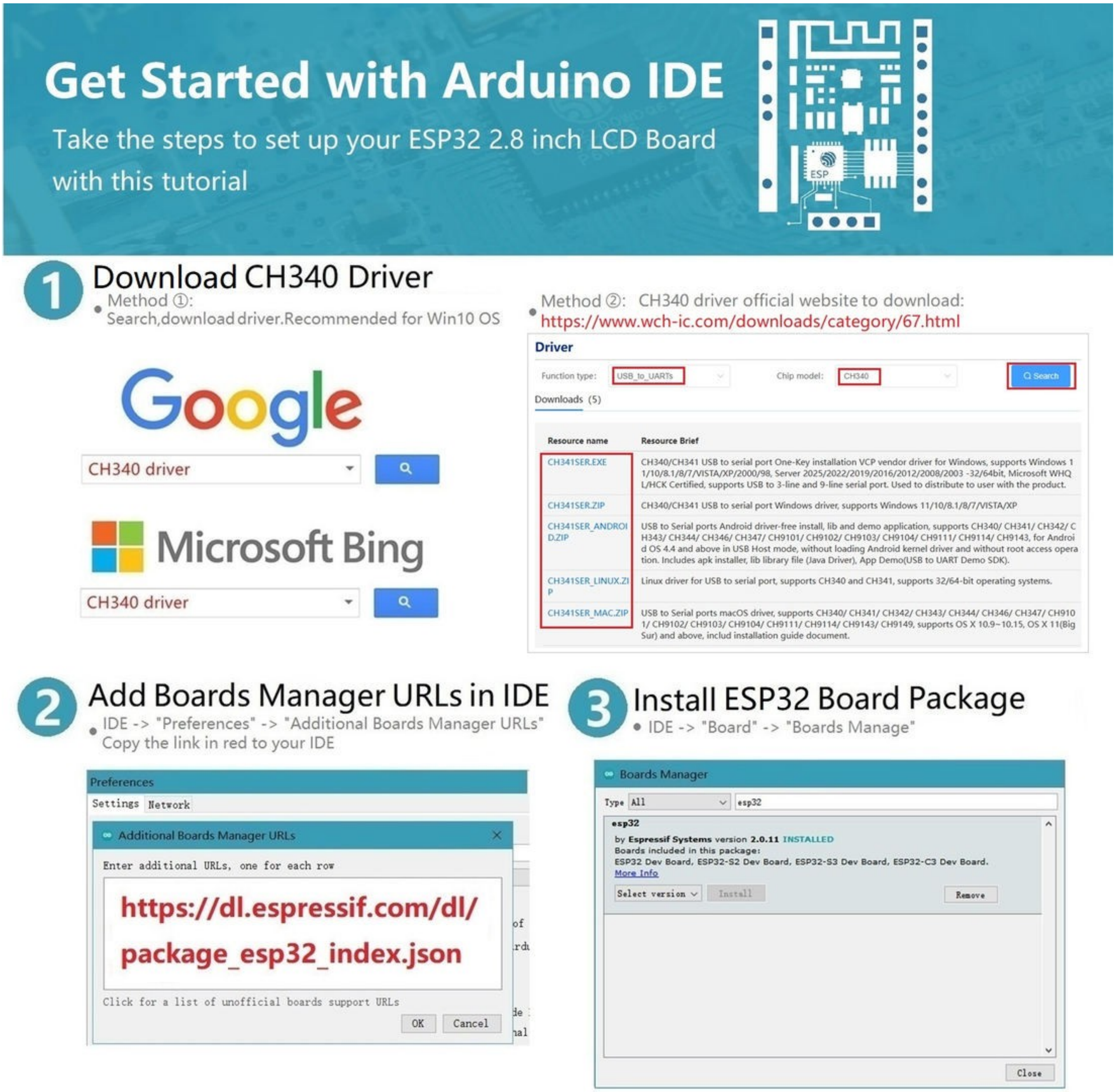
This guide takes you from an unopened box to a working display, touch panel and microSD card. Follow the four parts in order. Each part opens with the illustrated sheet, then explains every step in words.

The whole process takes about fifteen minutes the first time. Once the driver and the libraries are installed, you never need to repeat Parts 1 and 2.

What you need

- The ideaspark ESP32 2.8 inch LCD board
- A USB cable that carries data. Many charging cables carry power only and will not work.
- A computer running Windows, macOS or Linux
- The Arduino IDE, version 1.8.19 or Arduino IDE 2.x
- The stylus, the microSD card and the USB card reader supplied in the box. **The sample code and this guide are already on the card.**

Part 1 — Driver and board package



Steps 1 to 3 — driver download, boards manager URL, ESP32 package

Steps 1 to 3 install the USB serial driver so your computer can see the board, then add ESP32 support to the Arduino IDE. The written instructions below follow the sheet above.

1 Install the CH340 driver

The board uses a CH340 USB to serial chip. Windows, macOS and Linux all need this driver before the board will appear as a serial port. There are two ways to get it.

Method 1 — Search for it

Search for CH340 driver on Google or Bing and download the package for your operating system. This is the quickest route on Windows 10 and Windows 11.

Method 2 — Download it from the manufacturer

WCH publishes the driver on its own site. This is the safer route, because you always get the current official version rather than a repackaged copy.

```
https://www.wch-ic.com/downloads/category/67.html
```

On that page set Function type to USB_to_UARTs and Chip model to CH340, then click Search. Five downloads are listed:

File	Operating system
`CH341SER.EXE`	Windows — one click installer, recommended
`CH341SER.ZIP`	Windows — manual install
`CH341SER_MAC.ZIP`	macOS 10.9 and newer, including Big Sur and later
`CH341SER_LINUX.ZIP`	Linux, 32 and 64 bit
`CH341SER_ANDROID.ZIP`	Android 4.4 and newer, USB host mode

On Windows, CH341SER.EXE is the simplest option. Run it, click Install, then unplug and replug the board.

When the driver is working, a new COM port (Windows) or a new /dev/tty device (macOS and Linux) appears while the board is connected. If nothing appears, try a different USB cable before reinstalling the driver.

2 Add the ESP32 boards manager URL

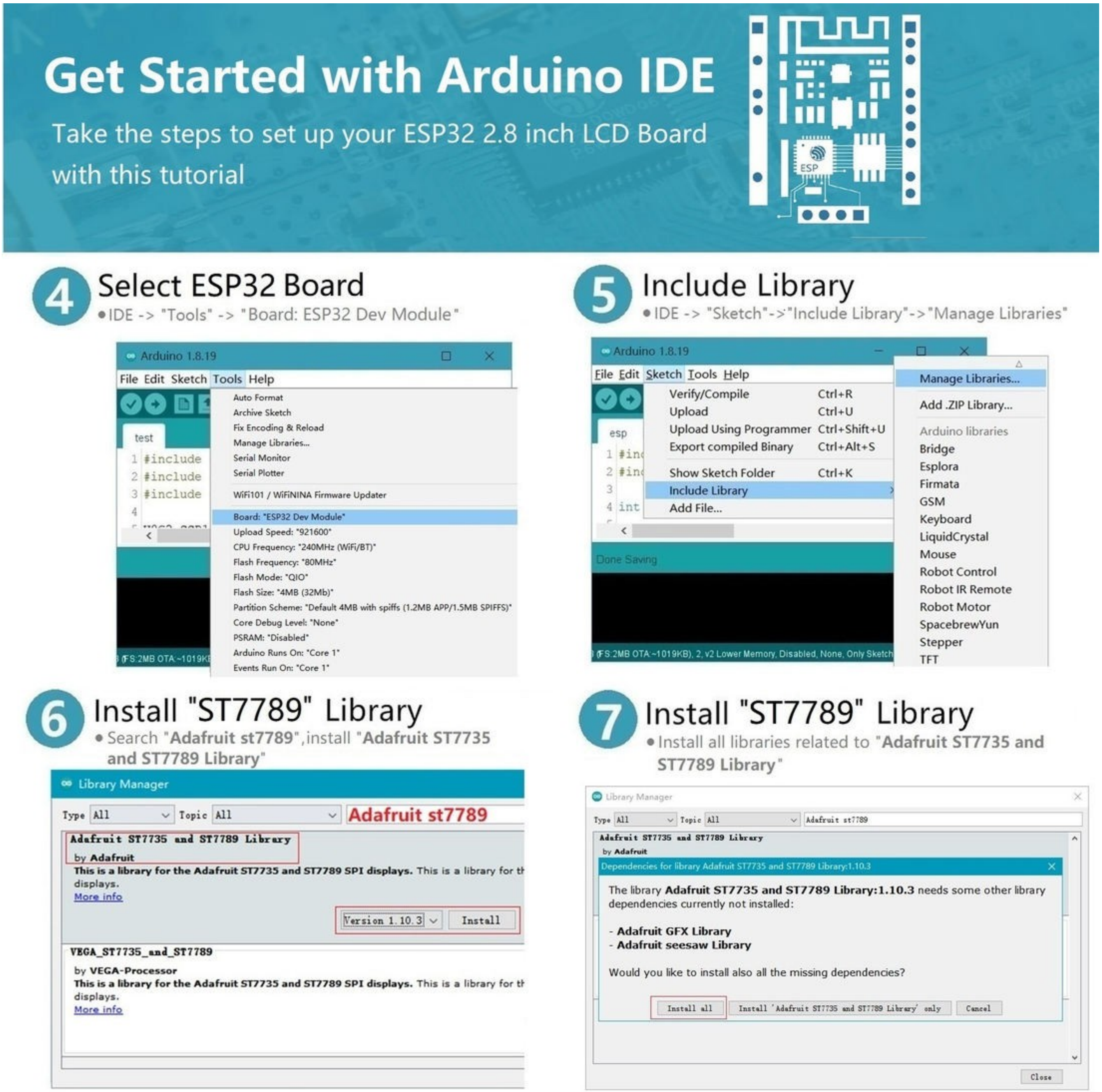
In the Arduino IDE open File, then Preferences, and find the field named Additional Boards Manager URLs. Paste this address into it and click OK.

```
https://dl.espressif.com/dl/package_esp32_index.json
```

3 Install the ESP32 board package

Open Tools, then Board, then Boards Manager. Search for esp32, select the package published by Espressif Systems and click Install. The download is large, so allow a few minutes.

Part 2 — Board selection and display libraries



Steps 4 to 7 — board selection and display libraries

Steps 4 to 7 point the IDE at the right board and install the two libraries that drive the screen.

4 Select the board

Open Tools, then Board, and choose ESP32 Dev Module. The full list of settings to use is given in Part 3.

5 Open the Library Manager

Open Sketch, then Include Library, then Manage Libraries.

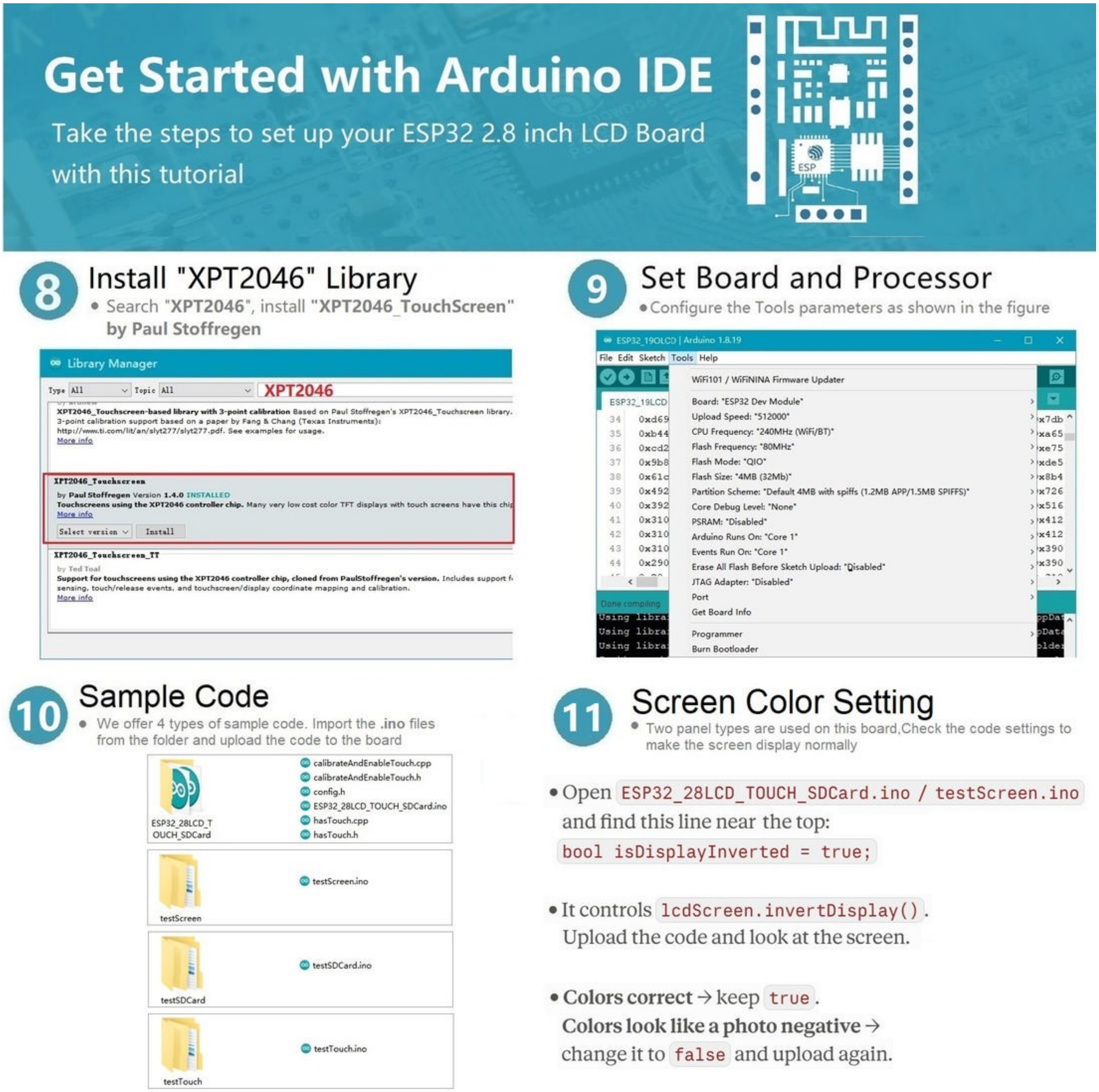
6 Install the display library

Search for Adafruit ST7789 and install Adafruit ST7735 and ST7789 Library, published by Adafruit.

7 Install its dependencies

The IDE will offer to install the libraries this one depends on. Choose Install all. Skipping this step is the most common reason the sketch fails to compile.

Part 3 — Touch library, settings and sample code



Steps 8 to 11 — touch library, board options, sample code, screen colour

Steps 8 to 11 add the touch library, set the upload options, load the sample code and set the screen colour option.

8 Install the touch library

Search for XPT2046 and install XPT2046_Touchscreen by Paul Stoffregen.

Version 1.4 or newer is required. Earlier versions do not accept an SPI argument and the sample code will not compile against them.

9 Set the board options

Configure the Tools menu as listed below.

Setting	Value
`Board`	ESP32 Dev Module
`Upload Speed`	921600 (drop to 512000 or 115200 if uploads fail)
`CPU Frequency`	240MHz (WiFi/BT)
`Flash Frequency`	80MHz
`Flash Mode`	QIO
`Flash Size`	4MB (32Mb)
`Partition Scheme`	Default 4MB with spiffs (1.2MB APP / 1.5MB SPIFFS)
`PSRAM`	Disabled
`Core Debug Level`	None
`Arduino Runs On`	Core 1
`Events Run On`	Core 1
`Port`	The COM port that appears when the board is plugged in

10 Load a sample sketch

Four sketches are supplied. Open the folder, then open the .ino file inside it and upload.

Sketch	What it does
`ESP32_28LCD_TOUCH_SDCard`	The full demo. Checks the display, the touch panel and the microSD card, then runs the touch calibration and a drawing screen.
`testScreen`	Display only. Use this first to confirm the screen works and to set the colour option in Part 4.
`testTouch`	Touch panel only. Tap with the stylus to draw.
`testSDCard`	microSD card only. Writes a file and reads it back. There is no screen output, so open the Serial Monitor at 115200 baud to see the result.

Start with testScreen. It is the quickest way to confirm the board is alive and it is where you set the colour option in Part 4.

11 Set the screen colour option

Two panel types are used on this board. One line of code selects the matching colour mode. See Part 4 for the full procedure.

Part 4 — Screen colour setting

Open ESP32_28LCD_TOUCH_SDCard.ino or testScreen.ino and find this line near the top of the file:

```
bool isDisplayInverted = true;
```

It controls lcdScreen.invertDisplay(). Upload the sketch and look at the screen.

- **Colours correct** — keep the value as `true`.
- **Colours look like a photo negative** — change it to `false` and upload again.

You only need to do this once. Use the same value in every sketch you upload to this board.

> **Reference point**
>
> On the boot screen, text should appear light on a dark background. If it appears dark on a light background, the setting is the wrong way round.

Part 5 — Touch calibration

The touch version ships with a five point calibration routine. The first time you run the full demo, the screen shows a series of targets.

1. Tap the screen once to begin.
2. Tap the centre of each target as it appears. There are five: four near the corners and one in the middle. Use the supplied stylus, not a fingertip.
3. When SAVED appears, the result is stored in flash and is kept through power cycles.
4. To calibrate again at any time, send the letter c over the Serial Monitor.

If the routine reports RETRY, the taps were too far from the targets. Take your time and press firmly enough for the panel to register a steady contact.

> **Resistive touch, not capacitive**
>
> This panel responds to pressure, so it works with a stylus, a fingernail or a gloved hand. A soft fingertip may not register reliably. That is normal for a resistive panel and is not a fault.

Part 6 — Adding your own hardware

The header layout matches the ESP32 DEVKIT V1, so pin numbers taken from existing ESP32 tutorials apply directly.

Signal	GPIO	Notes
SPI clock, data in, data out	`18, 19, 23`	Shared by the display, touch and microSD
Display CS / DC / RST / backlight	`15, 2, 4, 32`	Backlight must be driven HIGH
Touch CS / IRQ	`14, 27`	Touch version only
microSD CS	`13`	
Free for your project	`5, 12, 16, 17, 21, 22, 25, 26, 33`	Plus input-only 34, 35, 36, 39

Before you connect anything to GPIO12

GPIO12 is a strapping pin. If it is held HIGH while the board powers up, the ESP32 will not start and the board will appear dead.

Anything you connect there must leave the pin low or floating at power-up. A sensor module with a pull-up resistor on that line will stop the board booting. If this happens, disconnect it and power up again; no damage is done.

Analog inputs while Wi-Fi is running

The ESP32 cannot read its ADC2 block while Wi-Fi is active. For analog sensors on a connected board, use GPIO 33, 34, 35, 36 or 39.

Part 7 — Troubleshooting

Symptom	What to check
No COM port appears	The CH340 driver is not installed, or the cable is charge-only. Try a different USB cable first, then reinstall the driver from Step 1 and replug the board.
Upload fails or times out	Lower Upload Speed to 512000 or 115200. If it still fails, hold the BOOT button while the upload starts.
Screen stays black	The backlight pin must be driven HIGH. Confirm the sketch sets GPIO32 as an output and writes HIGH to it before drawing.
Colours look like a negative	Change isDisplayInverted as described in Part 4.
Touch is offset or ignores taps	Send the letter c over the Serial Monitor to run the calibration again. Use the stylus, not a fingertip.
microSD fails to initialise	Format the card as FAT32, and make sure the sketch sets the display and touch chip select pins HIGH before calling SD.begin(). A floating chip select corrupts the shared data line.
Sketch will not compile	Check the XPT2046_Touchscreen library is version 1.4 or newer. Older versions do not accept an SPI argument.
Board will not start after wiring	Disconnect anything attached to GPIO12 and power up again. See the warning in Part 6.
